

NFL INNOVATION HUB DASHBOARD - TECHNICAL INTEGRATION ARCHITECTURE

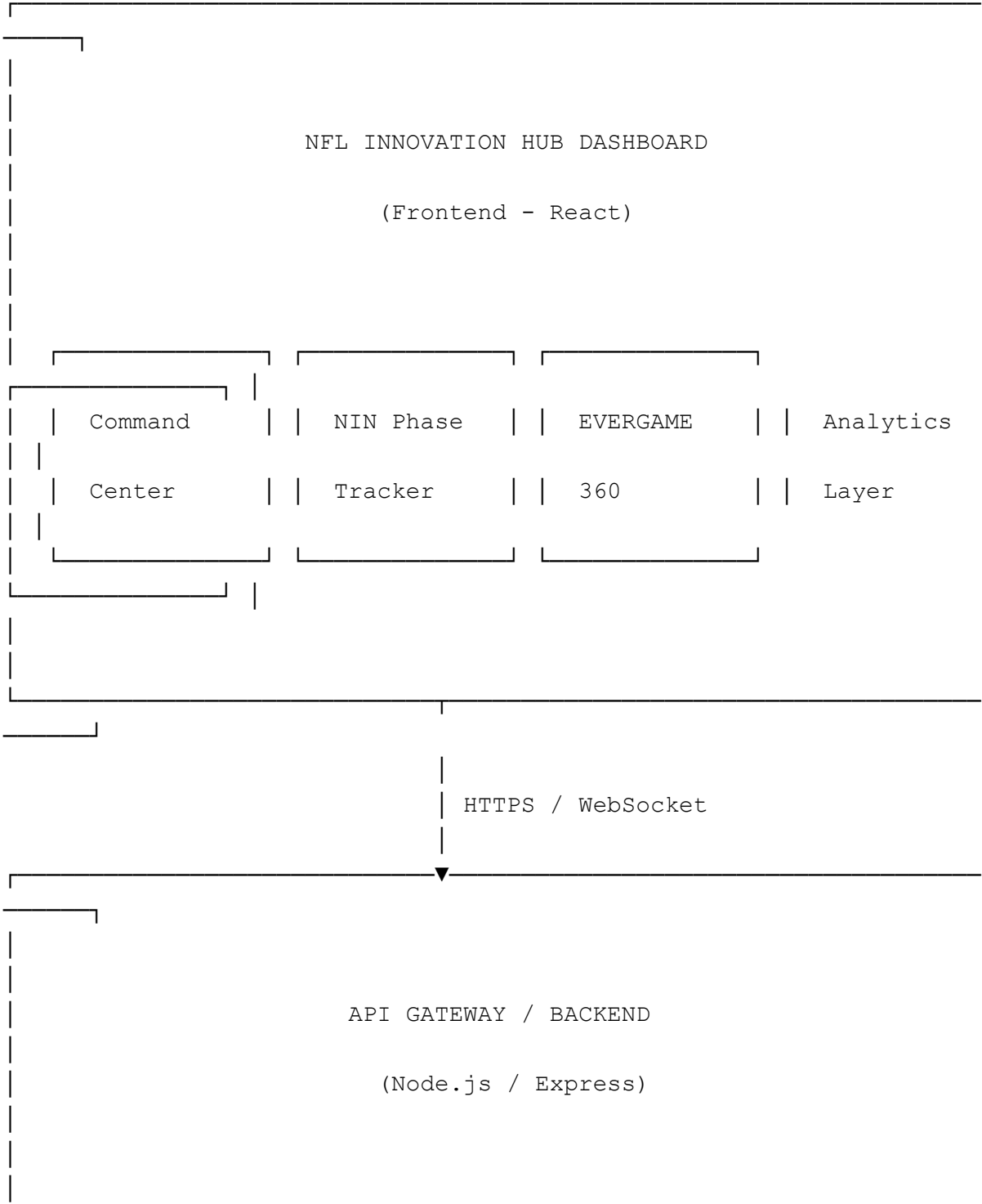
Version 1.0 | System Integration & Data Flow Specification

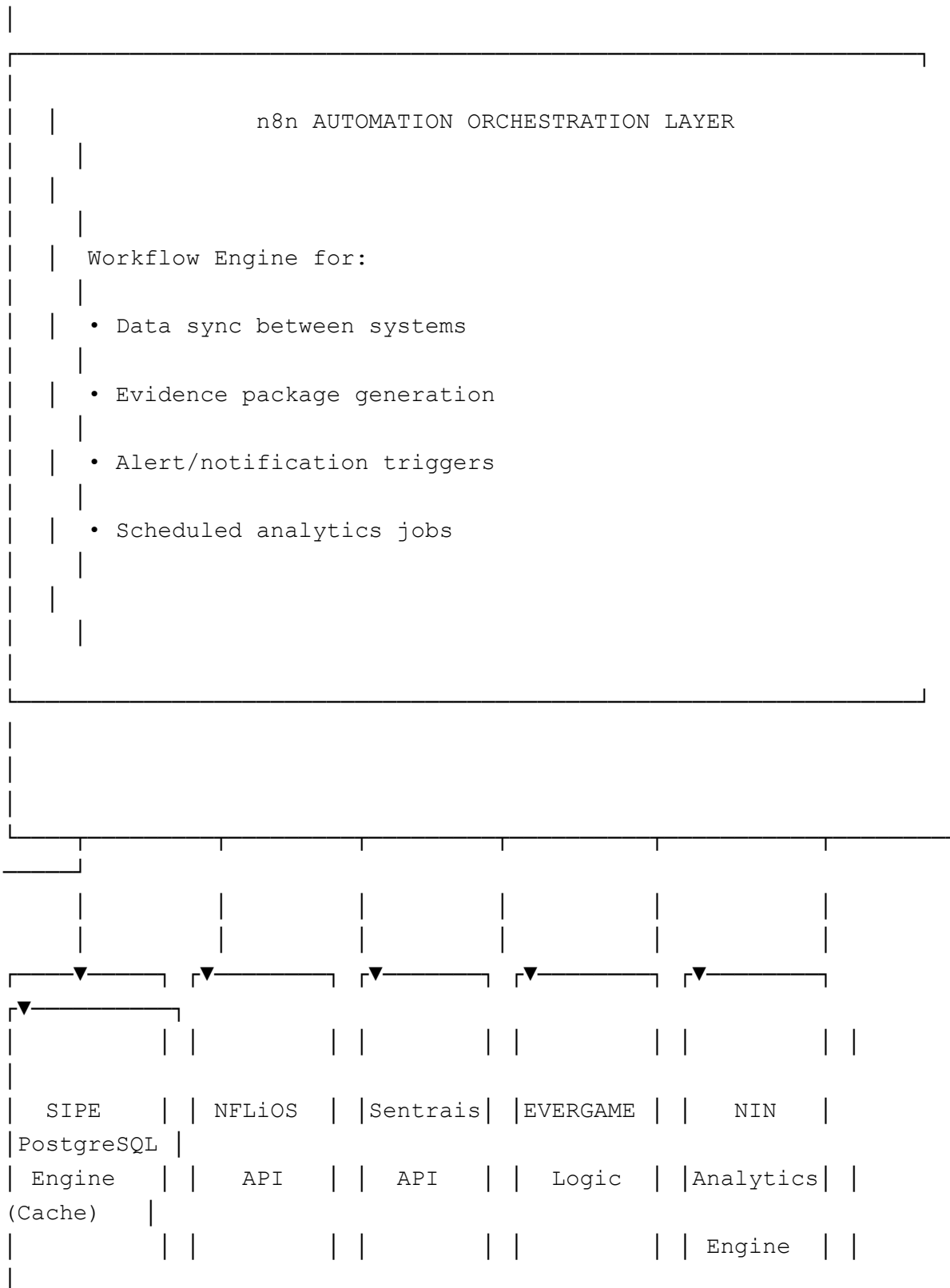
INTEGRATION OVERVIEW

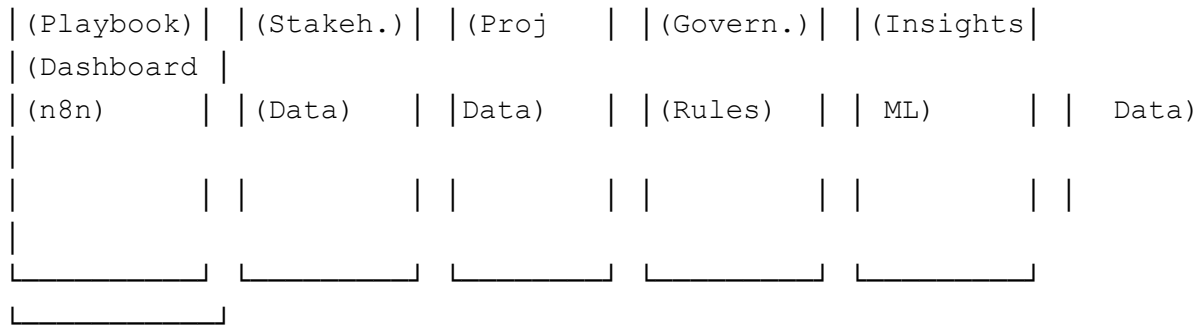
The NFL Innovation HUB Dashboard integrates five core systems into a unified intelligence platform:

1. **NOVATE Innovation Engine (SIPE)** - Automation & playbook execution
2. **NFLiOS** - NFL's operational intelligence & stakeholder data
3. **Sentraix Platform** - Project repository & workflow management
4. **EVERGAME 360** - Governance framework & decision logic
5. **NIN Analytics Engine** - Intelligence layer & insights

SYSTEM ARCHITECTURE DIAGRAM

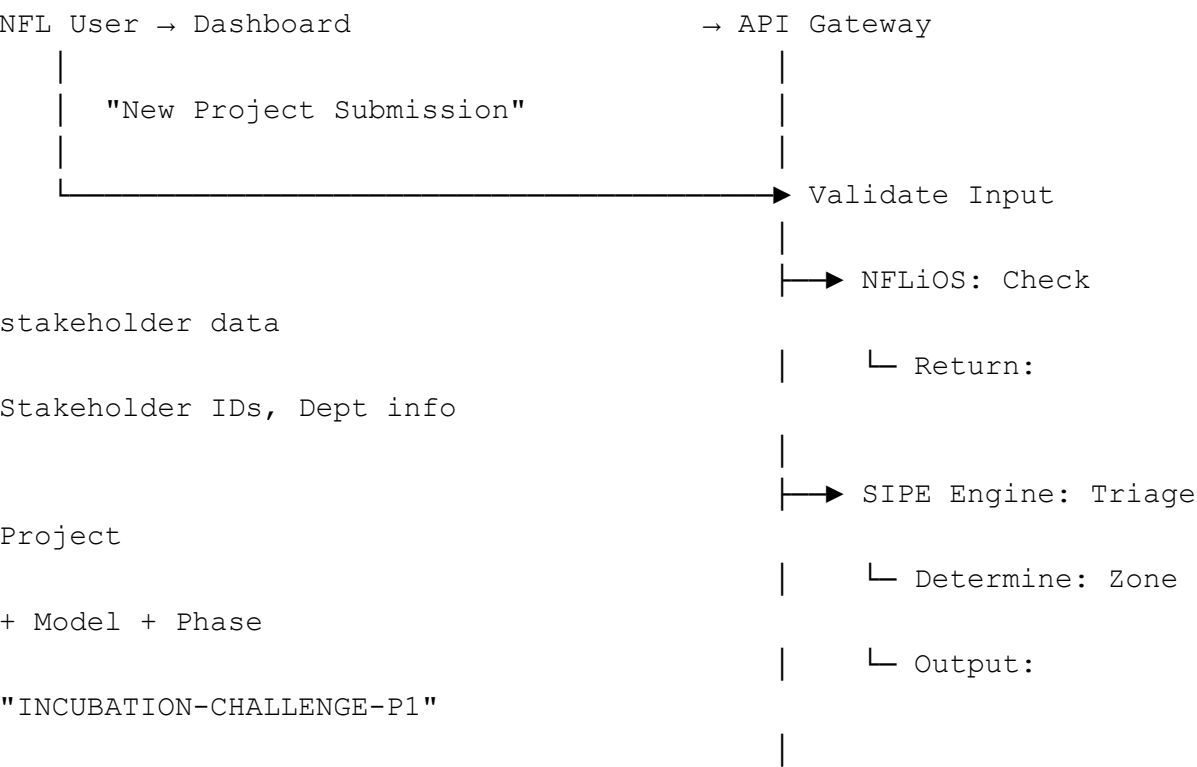


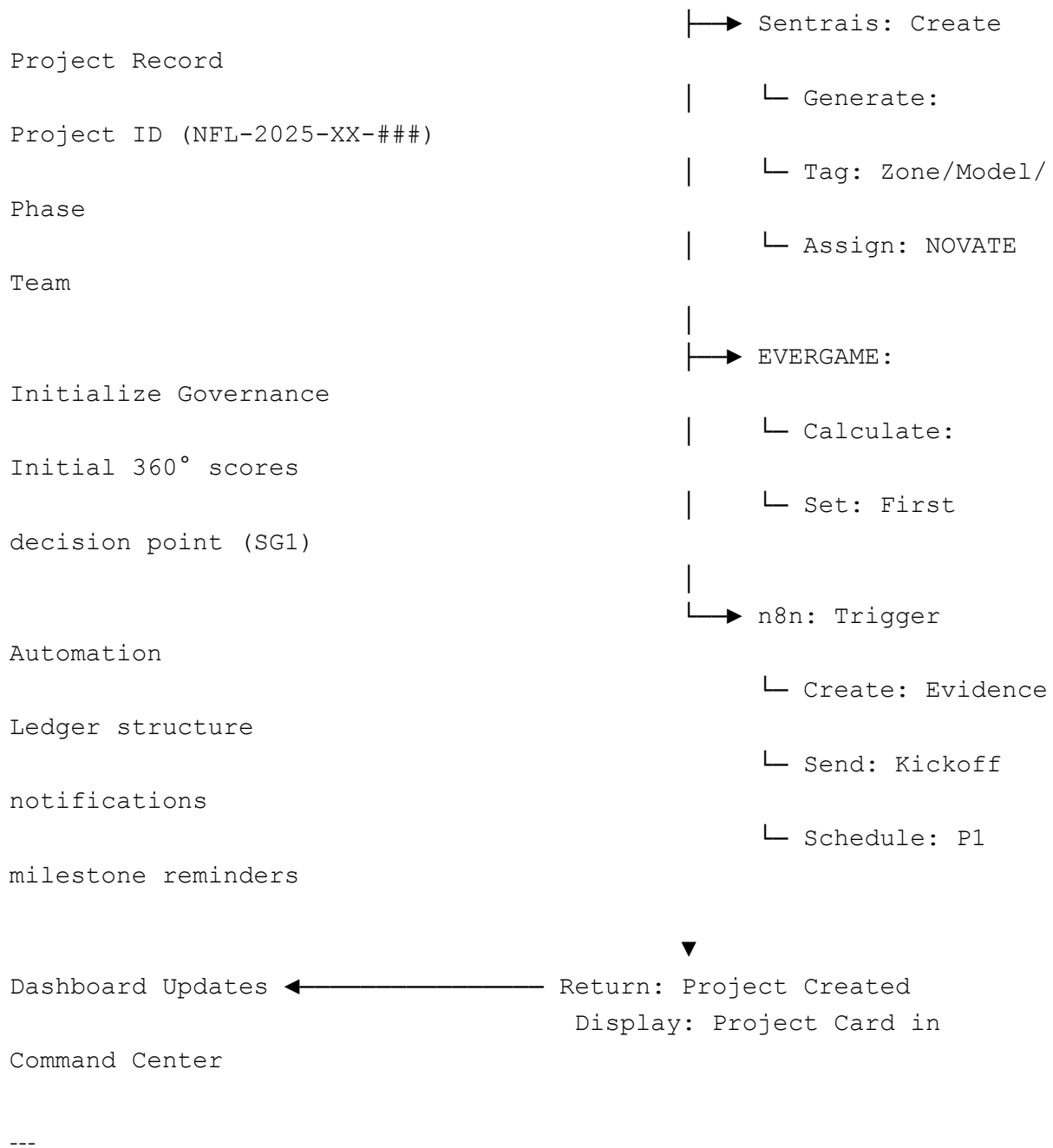




DATA FLOW SEQUENCES

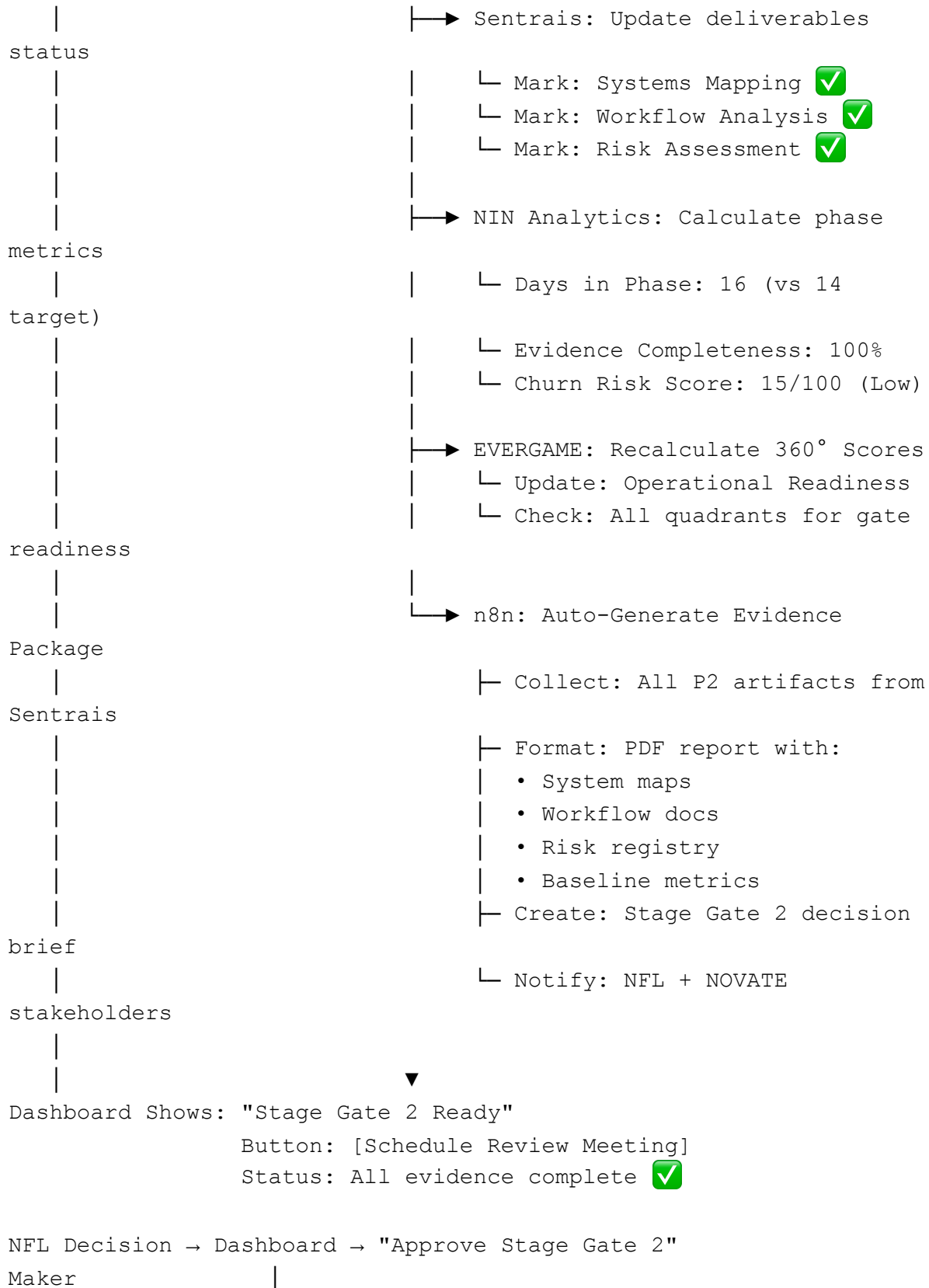
SEQUENCE 1: PROJECT CREATION & INTAKE

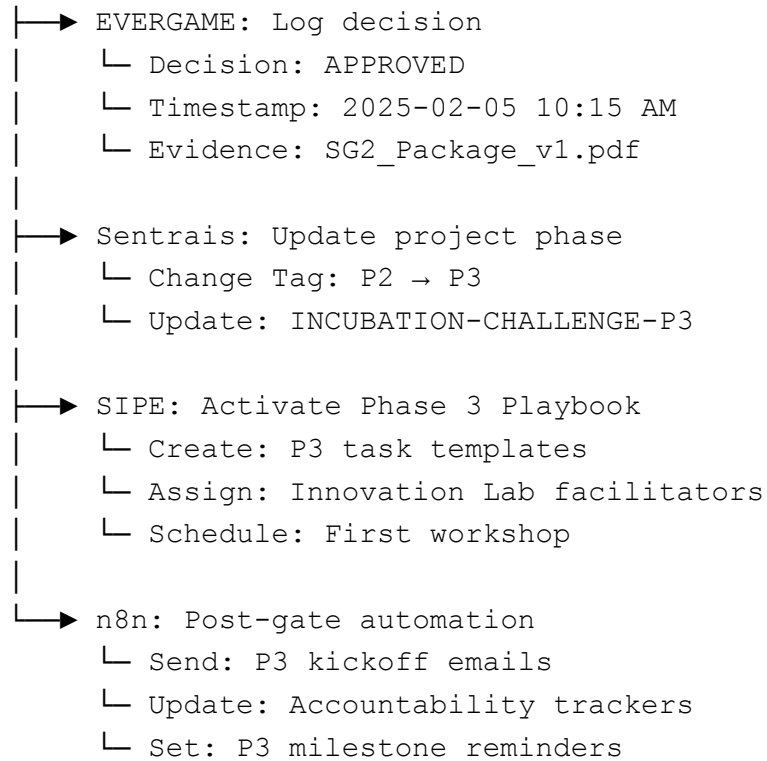




SEQUENCE 2: PHASE PROGRESSION & STAGE GATE

NOVATE PM → Dashboard → "Complete P2 Activities"

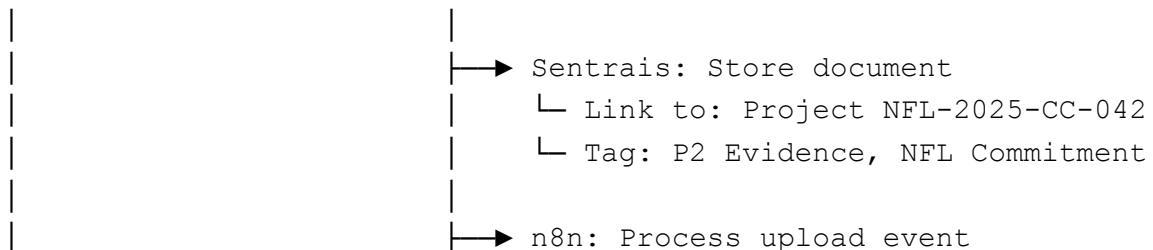


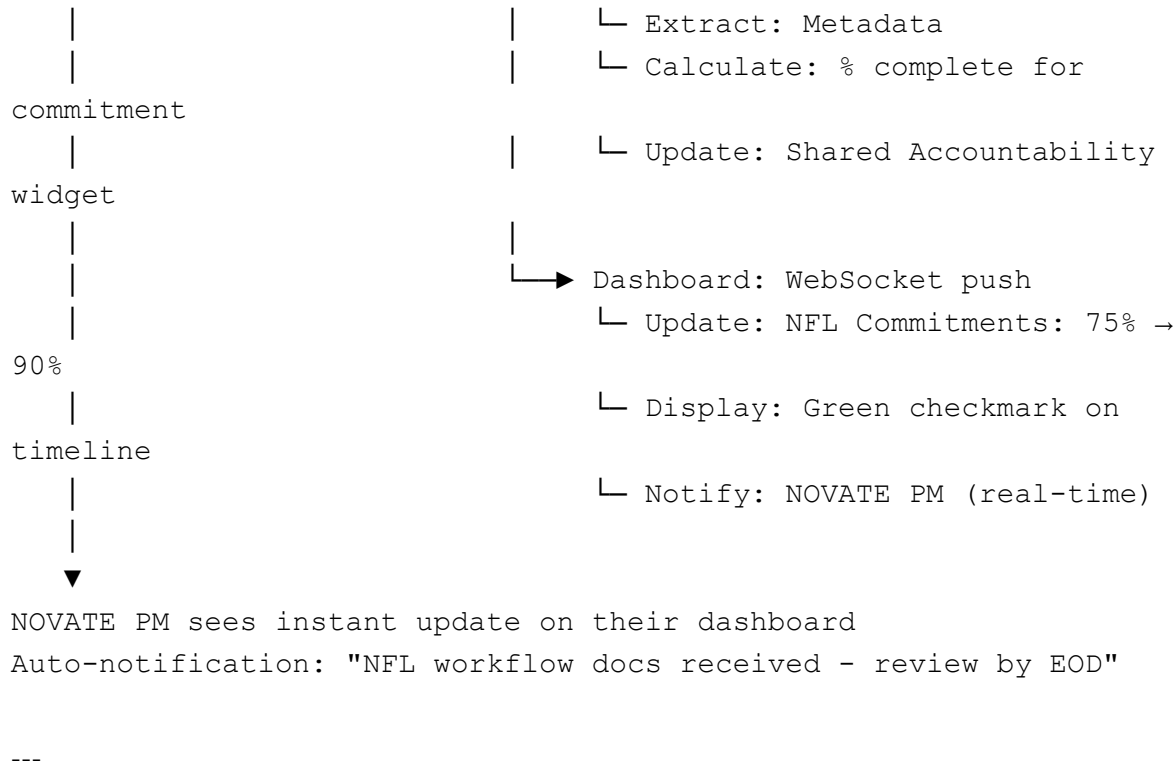


Dashboard Updates: Project now in Phase 3
New tasks visible
EVERGAME history updated

SEQUENCE 3: REAL-TIME ACCOUNTABILITY TRACKING

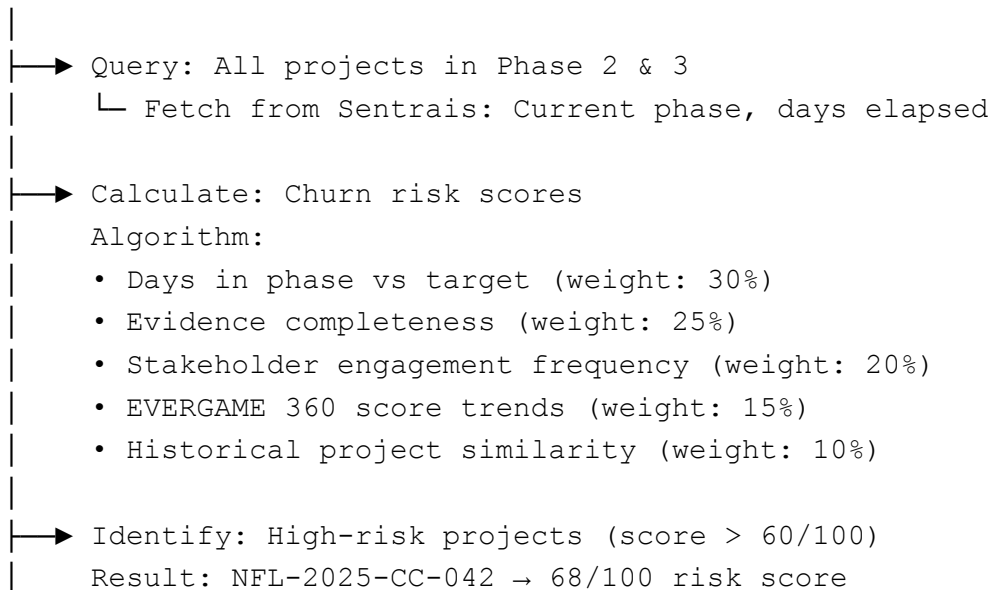
NFL User → Dashboard → "Upload Workflow Documentation"





SEQUENCE 4: CHURN RISK DETECTION & ALERT

NIN Analytics Engine (Scheduled: Daily 2 AM)




```

    | Reason: "P2 extended timeline + low stakeholder activity"
    |
    └─> n8n: Trigger alert workflow
        |
        └─> Dashboard: Add warning banner
            |
            | ⚠ "CHURN RISK: Phase 2 exceeding timeline by 4 days"
            |
            └─> Email: NOVATE Program Director
                |
                | Subject: "Churn Alert: Fan Experience AI (Risk:
68/100) "
                |
                | Body: Recommended interventions included
                |
                └─> Sentrais: Log alert in project timeline
                    |
                    | └─ Create: Auto-generated mitigation task
                    |
                    └─> EVERGAME: Flag in governance queue
                        |
                        | └─ Add: Early intervention decision point

```



Dashboard displays alert prominently

PM clicks "View Mitigation Options"

→ Shows pre-configured interventions from SIPE playbooks

API SPECIFICATIONS

Core Endpoints

1. Portfolio Summary

GET /api/v1/portfolio/summary

Response:

```
{
  "total_projects": 24,
  "portfolio_health": 82,
  "zones": {
    "incubation": { "count": 11, "percentage": 45 },
    "transformation": { "count": 7, "percentage": 29 },
    "performance": { "count": 4, "percentage": 17 },
    "productivity": { "count": 2, "percentage": 9 }
  },
  "models": {
    "challenge": 8,
    "cocreate": 10,
    "ops": 6
  },
  "attention_required": [
    {
      "type": "churn_risk",
      "count": 2,
      "project_ids": ["NFL-2025-CC-042", "NFL-2025-CH-018"]
    },
    {
      "type": "budget_variance",
      "count": 1,
      "project_ids": ["NFL-2025-OP-005"]
    }
  ]
}
```

```

    }
  ],
  "decision_points": [
    {
      "type": "stage_gate_2",
      "count": 3,
      "due_this_week": 2
    }
  ]
}

```

2. NIN Phase Detail

GET /api/v1/projects/:id/nin-phases

Response:

```

{
  "project_id": "NFL-2025-CC-042",
  "project_name": "Fan Experience AI Platform",
  "current_phase": 2,
  "phase_details": {
    "phase_1": {
      "name": "Discovery & Framing",
      "status": "completed",
      "start_date": "2025-01-02",
      "end_date": "2025-01-14",
      "days_elapsed": 12,
      "target_days": 14,
      "deliverables": [
        {
          "name": "Stakeholder Mapping",
          "status": "completed",
          "evidence_ledger_ids": ["EL-001", "EL-002"]
        }
      ],
      "stage_gate": {
        "number": 1,

```

```

    "name": "Discovery Approval",
    "date": "2025-01-14",
    "decision": "approved",
    "evidence_package_url": "/evidence/SG1_Package.pdf"
  }
},
"phase_2": {
  "name": "Blueprinting (NIN Forensics)",
  "status": "in_progress",
  "start_date": "2025-01-15",
  "days_elapsed": 18,
  "target_days": 21,
  "deliverables": [
    {
      "name": "Systems Mapping",
      "status": "completed",
      "owner_nfl": null,
      "owner_novate": "Sarah Chen",
      "evidence_ledger_ids": ["EL-005", "EL-006", "EL-007"]
    },
    {
      "name": "Workflow Analysis",
      "status": "in_progress",
      "progress_percentage": 80,
      "owner_nfl": "Mike Rodriguez",
      "owner_novate": "Sarah Chen",
      "due_date": "2025-01-30"
    }
  ],
  "churn_risk": {
    "score": 68,
    "level": "medium",
    "factors": [
      "Timeline extended by 4 days",
      "NFL stakeholder availability delayed"
    ],
    "mitigation": "Weekly sync added with NFL stakeholder"
  },
  "stage_gate": {

```

```

        "number": 2,
        "name": "Blueprint Validation",
        "scheduled_date": "2025-02-05",
        "readiness": 60,
        "decision_maker": "Jennifer Wu (NFL VP Innovation)"
    }
}
}
}

```

3. EVERGAME 360 Scores

GET /api/v1/projects/:id/evergame-scores

Response:

```

{
  "project_id": "NFL-2025-CC-042",
  "overall_score": 78,
  "quadrants": {
    "strategic_alignment": {
      "score": 85,
      "status": "green",
      "components": {
        "league_priority": { "score": 90, "weight": 40 },
        "sponsor_value": { "score": 85, "weight": 35 },
        "stakeholder_buyin": { "score": 80, "weight": 25 }
      }
    },
    "operational_readiness": {
      "score": 72,
      "status": "yellow",
      "components": {
        "resource_availability": { "score": 85, "weight": 30 },
        "technical_feasibility": { "score": 65, "weight": 40 },
        "timeline_viability": { "score": 70, "weight": 30 }
      }
    }
  },
}

```

```

    "risk_compliance": {
      "score": 91,
      "status": "green",
      "components": {
        "security_assessment": { "score": 95, "weight": 40 },
        "regulatory_compliance": { "score": 90, "weight": 35 },
        "interoperability_risk": { "score": 88, "weight": 25 }
      }
    },
    "financial_impact": {
      "score": 68,
      "status": "yellow",
      "components": {
        "budget_actual": { "score": 90, "weight": 30 },
        "projected_roi": { "score": 55, "weight": 45 },
        "cost_benefit": { "score": 70, "weight": 25 }
      }
    }
  },
  "shared_accountability": {
    "nfl_commitments": {
      "total": 4,
      "completed": 2,
      "in_progress": 1,
      "not_started": 1,
      "percentage": 75
    },
    "novate_deliverables": {
      "total": 5,
      "completed": 2,
      "in_progress": 1,
      "not_started": 2,
      "percentage": 60
    }
  }
}

```

4. Analytics - Phase Performance

GET /api/v1/analytics/phase-performance

Query Params:

- date_range: "last_12_months" | "last_quarter" | custom
- zone_filter: "incubation" | "transformation" | etc.
- model_filter: "challenge" | "cocreate" | "ops"

Response:

```
{
  "date_range": "last_12_months",
  "phase_cycle_times": [
    {
      "phase": 1,
      "name": "Discovery & Framing",
      "avg_days": 12.3,
      "target_days": 14,
      "variance_percentage": -12,
      "status": "green",
      "sample_size": 42
    },
    {
      "phase": 2,
      "name": "Blueprinting",
      "avg_days": 16.8,
      "target_days": 14,
      "variance_percentage": 20,
      "status": "yellow",
      "sample_size": 38,
      "insight": "NFL stakeholder alignment delays averaging 3.2
days",
      "recommendation": "Implement pre-Phase 2 stakeholder alignment
workshop"
    }
  ],
  "zone_migration": {
    "incubation_to_transformation": {
```

```

        "success_rate": 65,
        "successful": 13,
        "total": 20
    },
    "transformation_to_performance": {
        "success_rate": 80,
        "successful": 8,
        "total": 10
    }
},
"model_effectiveness": {
    "challenge": {
        "score": 78,
        "avg_time_to_pilot_days": 58,
        "success_rate": 62,
        "avg_roi": 2.1
    },
    "cocreate": {
        "score": 85,
        "avg_time_to_pilot_days": 72,
        "success_rate": 80,
        "avg_roi": 3.2
    }
}
}

```

5. Evidence Package Generation

POST /api/v1/evidence-package/generate

Request:

```

{
  "project_id": "NFL-2025-CC-042",
  "stage_gate": 2,
  "include_sections": [
    "phase_summary",
    "deliverables",

```



```
    "evidence_ledger",
    "risk_assessment",
    "evergame_scores",
    "recommendations"
  ]
}
```

Response:

```
{
  "package_id": "EP-2025-02-001",
  "project_id": "NFL-2025-CC-042",
  "stage_gate": 2,
  "generated_at": "2025-02-01T14:30:00Z",
  "generated_by": "Sarah Chen (NOVATE PM)",
  "status": "ready",
  "file_url": "/packages/EP-2025-02-001.pdf",
  "file_size_mb": 4.2,
  "sections": [
    {
      "name": "Phase 2 Summary",
      "pages": "1-3"
    },
    {
      "name": "Evidence Ledger",
      "pages": "4-12",
      "artifact_count": 8
    }
  ]
}
```

N8N AUTOMATION WORKFLOWS

Workflow 1: Project Intake & Triage

Workflow Name: "project_intake_triage"

Trigger: Webhook (NFL Dashboard → New Project Submission)

Steps:

1. Validate Input Data
 - Check required fields
 - Sanitize text inputs
2. NFLiOS Integration
 - GET /stakeholders/{id}
 - GET /departments/{id}
 - Enrich project data
3. SIPE Triage Logic
 - Evaluate: Project scope, department, complexity
 - Determine: Zone (Incubation/Transformation/etc.)
 - Determine: Model (Challenge/Co-Create/Ops)
 - Assign: Phase 1 (Discovery)
4. Sentrais Create Project
 - POST /projects
 - Generate: NFL-YYYY-[MODEL]-###
 - Tag: Zone-Model-Phase
5. EVERGAME Initialize
 - Calculate: Initial 360° scores
 - Create: First decision point (SG1)
6. Evidence Ledger Setup
 - Create: Folder structure in Sentrais
 - Generate: Phase 1 templates

7. Notifications

- Email: NOVATE Program Director
- Email: NFL Project Owner
- Slack: #novate-nfl-projects

8. Return Success

- Webhook response: Project created

Workflow 2: Churn Risk Monitoring

Workflow Name: "churn_risk_daily_scan"

Trigger: Schedule (Daily at 2:00 AM EST)

Steps:

1. Query Active Projects

- Sentra's API: GET /projects?status=active&phase=2,3

2. For Each Project:

a. Fetch Phase Data

- Days in current phase
- Evidence completeness %
- Stakeholder activity log

b. Fetch EVERGAME Scores

- Current 360° scores
- Score trend (last 7 days)

c. Calculate Churn Risk Score

- Algorithm: Weighted model
- Output: 0-100 risk score

3. Filter High Risk (Score > 60)

4. For Each High-Risk Project:

- a. Generate Alert

- **Create**: Dashboard notification
- **Log**: Sentrais project timeline
- b. Fetch Mitigation Playbook
 - **SIPE**: GET /playbooks/churn-mitigation
 - **Match**: Risk factors to interventions
- c. Create Auto-Task
 - **Sentrais**: POST /tasks
 - **Assign**: NOVATE PM
 - **Due**: Within 48 hours
- d. Send Notifications
 - **Email**: Program Director + PM
 - **Dashboard**: Real-time update
- 5. Generate Daily Report
 - **Summary**: All projects scanned
 - **Alerts**: High-risk projects
 - **Store**: PostgreSQL for trending

Workflow 3: Evidence Package Auto-Generation

Workflow Name: "evidence_package_generator"

Trigger: Manual (Dashboard button) or Scheduled (5 days before Stage Gate)

Steps:

1. Receive Input
 - Project ID
 - Stage Gate Number
 - Requested sections
2. Fetch Project Data (Sentrais)
 - Project metadata
 - Phase deliverables

- Evidence Ledger documents
3. Fetch EVERGAME Data
 - Current 360° scores
 - Decision history
 - Accountability status
 4. Fetch NIN Analytics
 - Phase performance vs target
 - Churn risk assessment
 - Recommendations
 5. Compile Document (Python/ReportLab)
 - **Template:** Stage Gate Evidence Package
 - **Sections:**
 - a. Executive Summary
 - b. Phase Completion Summary
 - c. Evidence Ledger (all artifacts)
 - d. Risk Assessment
 - e. EVERGAME 360 Scores
 - f. Shared Accountability Status
 - g. Recommendations for Next Phase
 6. Generate PDF
 - **Save:** Sentrais document library
 - **Name:** SG{number}_{project-id}_{date}.pdf
 7. Update Dashboard
 - **Link:** Evidence package available
 - **Notify:** NFL decision maker + NOVATE PM
 8. Schedule Meeting (Optional)
 - Google Calendar API
 - **Create:** Stage Gate review meeting
 - **Invite:** Stakeholders
 - **Attach:** Evidence package

DATA MODELS

Project Schema (Sentrais)

```
interface Project {
  id: string; // NFL-2025-CC-042
  name: string;
  description: string;

  // Zone-Model-Phase Taxonomy
  zone: 'incubation' | 'transformation' | 'performance' |
'productivity';
  model: 'challenge' | 'cocreate' | 'ops';
  current_phase: 1 | 2 | 3 | 4 | 5 | 6;
  phase_tag: string; // "TRANSFORMATION-COCREATE-P2"

  // Dates
  created_at: DateTime;
  phase_started_at: DateTime;
  target_phase_end: DateTime;

  // Stakeholders
  nfl_owner: {
    id: string;
    name: string;
    role: string;
    department: string;
  };
  novate_pm: {
    id: string;
    name: string;
  };

  // Budget
  budget_allocated: number;
```

```

budget_spent: number;

// Status
status: 'active' | 'paused' | 'completed' | 'cancelled';
churn_risk_score: number; // 0-100

// Evidence
evidence_ledger: EvidenceLedgerEntry[];

// Governance
stage_gates: StageGate[];
}

```

EVERGAME Score Schema

```

interface EVERGAMEScore {
  project_id: string;
  calculated_at: DateTime;

  overall_score: number; // 0-100

  quadrants: {
    strategic_alignment: QuadrantScore;
    operational_readiness: QuadrantScore;
    risk_compliance: QuadrantScore;
    financial_impact: QuadrantScore;
  };

  shared_accountability: {
    nfl_commitments: AccountabilityMetric;
    novate_deliverables: AccountabilityMetric;
  };
}

interface QuadrantScore {
  score: number; // 0-100
}

```

```

status: 'green' | 'yellow' | 'red';
components: {
  [key: string]: {
    score: number;
    weight: number; // percentage
  };
};
}

```

Evidence Ledger Entry Schema

```

interface EvidenceLedgerEntry {
  id: string; // EL-001
  project_id: string;
  phase: number;

  deliverable_name: string;
  artifact_type: 'document' | 'presentation' | 'spreadsheet' |
'code' | 'other';

  file_url: string;
  file_size_bytes: number;

  created_by: {
    name: string;
    organization: 'NFL' | 'NOVATE';
  };
  created_at: DateTime;

  tags: string[];
  description: string;
}

```

SECURITY & ACCESS CONTROL

Role-Based Permissions

ROLE: NFL_ADMIN

- |─ Dashboard Access: ALL views
- |─ Project Visibility: ALL projects
- |─ Actions:
 - | |─ Create new projects
 - | |─ Approve stage gates
 - | |─ View Evidence Ledger
 - | |─ Export reports
 - | |─ Manage NFL user accounts

ROLE: NFL_PROJECT_OWNER

- |─ Dashboard Access: ALL views
- |─ Project Visibility: ASSIGNED projects only
- |─ Actions:
 - | |─ Upload evidence documents
 - | |─ Update commitment status
 - | |─ View EVERGAME scores
 - | |─ Export project reports

ROLE: NOVATE_PROGRAM_DIRECTOR

- |─ Dashboard Access: ALL views
- |─ Project Visibility: ALL projects
- |─ Actions:
 - | |─ Create/edit projects
 - | |─ Manage team assignments
 - | |─ Override churn risk status
 - | |─ Access NIN Analytics (all features)
 - | |─ System configuration

ROLE: NOVATE_PM

- |─ Dashboard Access: Command Center, NIN Tracker, EVERGAME

- | Project Visibility: ASSIGNED projects
- | Actions:
 - | | Update phase deliverables
 - | | Upload evidence
 - | | Generate evidence packages
 - | | View analytics for assigned projects

ROLE: NOVATE_ARCHITECTURE

- | Dashboard Access: Analytics, System Monitoring
- | Project Visibility: AGGREGATED data only (no individual details)
- | Actions:
 - | | View system performance
 - | | Access efficiency recommendations
 - | | Monitor API health
 - | | PROTECTED: Cannot view NIN methodology internals

Data Protection

Core Secrets Protected from Client View:

- SIPE playbook automation logic
- NIN scoring algorithms
- Partner network/sourcing methods
- NOVATE internal cost structures
- Proprietary assessment templates

Data Encryption:

- TLS 1.3 for all data in transit
- AES-256 for sensitive data at rest
- JWT tokens for authentication (1-hour expiry)

- API keys rotated monthly

Audit Logging:

- All user actions logged with timestamp
- Evidence package access tracked
- Stage gate decisions recorded immutably
- Data exports logged and watermarked

PERFORMANCE & SCALABILITY

Caching Strategy

L1: Browser Cache (React Query)

- └ Portfolio summary: 5 minutes
- └ Project list: 5 minutes
- └ User preferences: Session duration

L2: API Gateway Cache (Redis)

- └ EVERGAME scores: 10 minutes
- └ Analytics aggregates: 1 hour
- └ Phase templates: 24 hours

L3: Database Query Cache

- └ Complex analytics: Pre-computed nightly
- └ Historical trends: Materialized views

Load Handling

Current Capacity (POC):

- Concurrent users: 50
- Projects tracked: 500
- API requests/sec: 100

Scaling Plan (Production):

- Concurrent users: 200+
- Projects tracked: 2000+
- API requests/sec: 500+
- Strategy: Horizontal scaling (Kubernetes)

Monitoring & Alerts

System Health Checks (Every 1 min):

- └ API Gateway: Response time < 500ms
- └ Database: Connection pool availability
- └ Sentrais API: Uptime check
- └ NFLiOS API: Uptime check
- └ n8n: Workflow queue length

Alert Thresholds:

- └  Warning: Response time > 1s
- └  Warning: Error rate > 1%
- └  Critical: Response time > 3s
- └  Critical: Error rate > 5%
- └  Critical: Any system unavailable > 2 min

Notification Channels:

- | Email: NOVATE Architecture Team
- | Slack: #novate-system-alerts
- | PagerDuty: On-call engineer (for critical)

DEPLOYMENT ARCHITECTURE

POC Environment

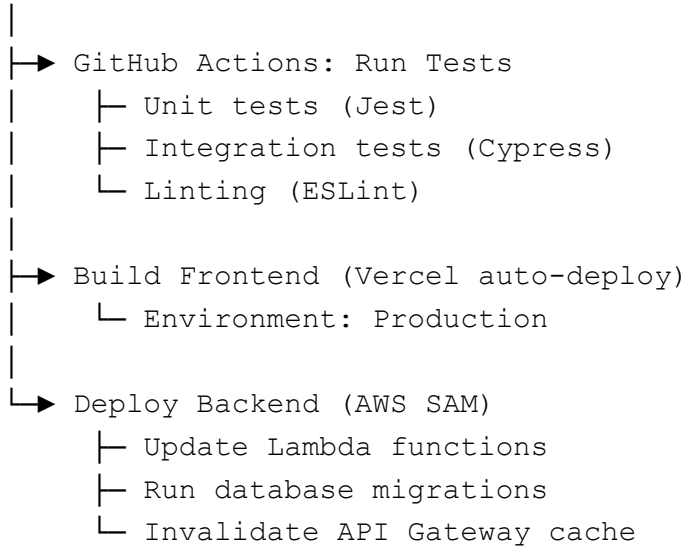
Cloud Provider: AWS / Google Cloud
Region: US-East (for NFL compliance)

Services:

- | Frontend: Vercel (React app)
 - | | CDN: Cloudflare
- | Backend API: AWS Lambda + API Gateway
 - | | Language: Node.js 18
- | Database: PostgreSQL (AWS RDS)
 - | | Size: db.t3.medium (POC)
- | Cache: Redis (AWS ElastiCache)
 - | | Size: cache.t3.small
- | Automation: n8n (Self-hosted on EC2)
 - | | Instance: t3.large
- | File Storage: AWS S3
 - | | Evidence packages, exports

CI/CD Pipeline

Code Push → GitHub



Deployment Frequency: On merge to main branch

Rollback Strategy: Instant revert to previous version

TESTING STRATEGY

Unit Tests

- **Frontend Components:** 80% coverage minimum
- **API Endpoints:** 90% coverage minimum
- **Business Logic:** 95% coverage minimum

Integration Tests

- **Critical Flows:**
- Project creation → Sentrais sync
- Phase progression → EVERGAME update
- Evidence package generation
- Real-time dashboard updates

E2E Tests (Cypress)

- **User Journeys:**
- NFL Admin creates project
- NOVATE PM progresses through phases
- Evidence package generation
- Stage gate approval
- Dashboard updates reflect all changes

Load Testing

- **Scenarios:**
- 50 concurrent users browsing dashboard
- 10 simultaneous evidence package generations
- Real-time updates to 100 projects
- **Tool:** Apache JMeter

- **Target:** <2s response time at 50 users

VERSION CONTROL & CHANGELOG

Document Version: 1.0

Related Documents:

- NFL_Innovation_HUB_Dashboard_POC_v1.0.md
- NFL_Innovation_HUB_Dashboard_Wireframes_v1.0.md

Date: November 28, 2025

Status: Technical Specification - Ready for Implementation

Change Log:

- v1.0 (Nov 28, 2025): Initial technical architecture
- All integration points defined
- API specifications complete
- n8n workflows documented
- Security model finalized

Next Steps:

1. Week 1: Set up development environment
2. Week 2: Implement core API endpoints
3. Week 3: Build n8n automation workflows
4. Week 4: Integrate with Sentrais/NFLiOS (test APIs)

5. Week 5-8: Frontend development + testing
6. Week 9: POC deployment & NFL testing

END OF TECHNICAL INTEGRATION ARCHITECTURE

For implementation questions, contact NOVATE Labs Technical Architecture Team.